9.3 Collecting Data

Name _____ Period ____ Date _____

Algebra II 9.3 (30-minute plan) · Student sheet

I can…

- identify types of samples and tell whether a sample is biased.
- identify the method of data collection.
- recognize biased survey questions and rewrite them.

Vocabulary · fill in the missing word

Random sample — every member of the _____ has an equal chance of being chosen

Self-selected sample — members _____

Systematic sample — a rule is used, such as every 10th _____

Stratified sample — the _____ is divided into groups, and members are randomly chosen from each group

Cluster sample — the _____ is divided into groups, and all members of some randomly chosen groups are surveyed

Convenience sample — only _____ who are easy to reach are chosen

Bias — a sample or question that overrepresents or _____ part of the population

Methods — experiment, _____ study, survey, simulation

Part 1 · Notes and examples (fill in each note, then work the example)

Stratified: random members from _____ group. **Cluster:** _____ members of some groups.

Example 1

Identify the type of sample.

a. every 10th student who enters the gym

b. all students in 3 randomly chosen homerooms

c. 20 randomly chosen students from each grade

d. students who answer a poll on the school website

Example 2

A town wants to know whether residents support building a new park. It surveys people at an existing park. Is the sample biased? Explain.

Methods of collecting data: Experiment: apply a _____. Observational study: _____ without changing anything.

Biased questions: A biased question _____ people toward an answer.

Example 3

Identify the method of data collection.

a. A researcher records how many cars run a red light at an intersection.

b. A computer flips 1,000 virtual coins.

c. Doctors give half of the patients a new drug.

d. A questionnaire is mailed to customers.

Part 2 · Partner practice

Partner practice

Identify the type of sample.

a. A coach randomly picks 2 players from each grade.

b. A teacher asks the first 10 students who arrive.